

Qualifying Exam in Microeconomics
Arizona State University
June 2024

There are four questions and you have four hours to complete the exam. If you believe that a question is in error, please explain the error and continue working.

1. Sylvia lives for $T > 1$ dates and consumes a single physical good at each date. Her consumption set is $X = \mathbb{R}_+^T$, where $x_t \geq 0$ is her consumption at date t , $t = 1, \dots, T$. Her preferences on X are represented by a *continuous* and *locally nonsatiated* function u on X . Her associated indirect utility function is

$$V(p, p \cdot \omega) = \max_{x \in B(p, p \cdot \omega)} u(x),$$

where $p \in \mathbb{R}_{++}^T$ is a vector of prices, $\omega \in \mathbb{R}_{++}^T$ is her endowment of date-contingent consumption, and $B(p, w) = \{x \in X \mid p \cdot x \leq w\}$ is her budget set at price-wealth pair (p, w) .

Consider this problem, call it *Problem I*:

Problem I: Choose an endowment ω out of a nonempty, compact constraint set $C \subset \mathbb{R}_{++}^T$, to maximize $V(p, p \cdot \omega)$.

In what follows you should invoke what you know about an indirect utility function under the given assumptions. (One approach is to read parts (a) and (b) before attempting an answer; the parts can be answered simultaneously.)

- (a) If the price of date-1 consumption, p_1 , increases, then what happens to her choice of ω_1 in Problem I: does it (i) *not increase*, (ii) *not decrease*, or is it (iii) impossible to say from the given facts? If it is impossible to say, explain why; if it is (i) or (ii), then say which, and prove your assertion.
- (b) For $t = 0, 1$, let ω^t solve Problem I for $p^t \gg 0$ with $p^1 \neq p^0$. What, if anything, can you say about the relationship between ω^0 and ω^1 ? Prove what you can say. (Whatever you can say will of course depend on p^1 and p^0 .)
- (c) Taking a step back, implicit in the formulation of Problem I is this sequential problem: first choose $\omega \in C$; given that ω choice, then choose $x \in B(p, p \cdot \omega)$. Suppose instead that Sylvia chooses ω and x *simultaneously*. Call this *Problem II*:

Problem II: Choose $(\omega, x) \in C \times X$ to maximize $u(x)$ subject to the constraint that $x \in B(p, p \cdot \omega)$.

Articulate your understanding of the relationship between the set of solutions to Problem I and the set of solutions to Problem II. We do not ask you to prove anything; just write as precisely as you can the relationship between the two sets of solutions.

- (d) Part (c) compared the solution sets to Problems I and II. Do the assumptions made ensure that Problem I has a solution? Why or why not?

2. Consider the two-player game with payoff matrix indicated below where β is a number close to 1.

	D	E	F
A	$0, 0$	$1, \beta$	$\beta, 1$
B	$\beta, 1$	$0, 0$	$1, \beta$
C	$1, \beta$	$\beta, 1$	$0, 0$

- (a) Find the unique (possibly-mixed) Nash equilibrium of the game when β is close to 1 but $\beta \neq 1$.
- (b) Find all mixed-strategy Nash equilibria when $\beta = 1$.
- (c) Is there a sense in which the set of mixed Nash equilibria is continuous in the parameter β at $\beta = 1$? Rephrasing formally, is there a sense in which the *Nash equilibrium correspondence*—mapping values of β in a small neighborhood of $\beta = 1$ to the set of Nash equilibria— is continuous at $\beta = 1$? Explain.

3. Consider a game with two players, a salesman of a good and a buyer.¹ The price of the good is $p > 0$ dollars per unit, and it is beyond the control of either player.

The buyer's preferences over the good and quality are represented by a quasilinear vN-M utility

$$u(q, \theta) = \theta q - \frac{1}{2}q^2 - pq,$$

where $q \geq 0$ is the quantity she buys and $\theta \in \mathbb{R}$ is a parameter representing the good's quality. The quality parameter has two possible values, θ_L and θ_H , where

$$0 < \theta_L < \theta_H \text{ and } \theta_L > p,$$

and the buyer initially believes that the two quality levels are *equally likely*.

The salesman earns a sales commission given by a continuous, strictly increasing function $R : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ of the quantity q bought. In what follows, $R(\cdot)$ is the salesman's vN-M utility.

Now to the game. First, nature chooses $\theta \in \{\theta_L, \theta_H\}$ with *equal chances* and reveals it to the salesman, but not to the buyer. Second, the type- t salesman chooses a message m out of a set $M_t = \{m_0, m_t\}$ to send to the buyer. For $t = L, H$, the message m_t is "the quality of the good is t ." The message m_0 is "the quality is either L or H ."

Note that the salesman does not have to tell the whole truth, but cannot lie: for $t = L, H$, a type- t salesman can only send the message m_t or the message m_0 .

After observing the salesman's message, the buyer decides how many units, if any, to buy at price p . Then the game ends.

- (a) As a preliminary, derive the buyer's demand for the good as a function of its price p and her belief about its quality. Carefully justify your derivation. (I suggest the notation $\mathbb{E}[\theta \mid \pi]$ to denote the expected quality conditional on the buyer's belief π that the quality is low.)
- (b) Identify the strategy set of each player. Define precisely the notion of a Bayesian-Nash equilibrium (BNE) for this game. Is there a BNE in which both salesman types send the same message? Explain.
- (c) (i) Define precisely the notion of a weak perfect Bayesian equilibrium (wPBE) for this game. (ii) Identify every wPBE and explain why each is a wPBE and why there are no other. (iii) Is there a wPBE in which *some* salesman type chooses $m = m_0$? (iv) Is there a wPBE in which the buyer does not learn the good's quality?
- (d) Now suppose that communication is impossible: each seller type has only one possible message, m_0 , so the buyer must decide how much of the good to buy without learning anything from the salesman. Compare the *ex ante* (before-Nature's-move) equilibrium welfare of each player in the (unique) wPBE of this no-communication game to welfare in every wPBE you identified in (c). Is it possible the salesman *ex ante* prefers the equilibrium of the no-communication game to some equilibria in (c)? If so, then under what assumptions on R ? How about the buyer?

¹Since the salesperson is keen to lie whenever possible, the salesperson is a he; the wily buyer is a she.

4. Consider an independent private values model (IPVM) with I bidders indexed by $i = 1, \dots, I$. Each bidder i draws her private valuation or type $x_i \in X_i = [a_i, b_i]$, $0 < a_i < b_i \in \mathbb{R}$ according to a cumulative distribution function F_i with a continuous and strictly positive density function f_i .

- (a) Define a direct revelation mechanism (DRM) for the model described.
- (b) Let $I = 1$ and $F_1(x_1) = x_1^2$ for $x_1 \in [0, 1]$. Describe the revenue-maximizing DRM; justify your answer.
- (c) Let $I = 1$ and $x_1 \in [0, 1]$. For each utility function given below, indicate whether it corresponds to the utility of a DRM. Either provide the corresponding DRM or explain why it does not exist.

i.

$$U_1(x_1) = \frac{1}{2}x_1^2.$$

ii.

$$U_1(x_1) = \frac{2}{3}x_1^2.$$

iii.

$$U_1(x_1) = \begin{cases} 0 & \text{if } x_1 \in [0, 1/8] \\ \frac{x_1}{3} - \frac{1}{24} & \text{if } x_1 \in [1/8, 1/2] \\ \frac{x_1}{2} - \frac{1}{8} & \text{if } x_1 \in [1/2, 3/4] \\ x_1 - \frac{1}{2} & \text{if } x_1 \in [3/4, 1] \end{cases}$$

iv.

$$U_1(x_1) = 2x_1^2 - \frac{3}{2}x_{\textcolor{red}{1}}^2 + \frac{1}{2}x_{\textcolor{red}{1}}$$

- (d) Let $I = 2$, $X_1 = [0, 1]$, $F_1(x_1) = x_1^2$; $X_2 = [0, 2]$, $F_2(x_2) = \frac{x_2}{2}$.

Find the DRM that maximizes seller's expected revenue and the DRM that maximizes social surplus. Justify your answers.